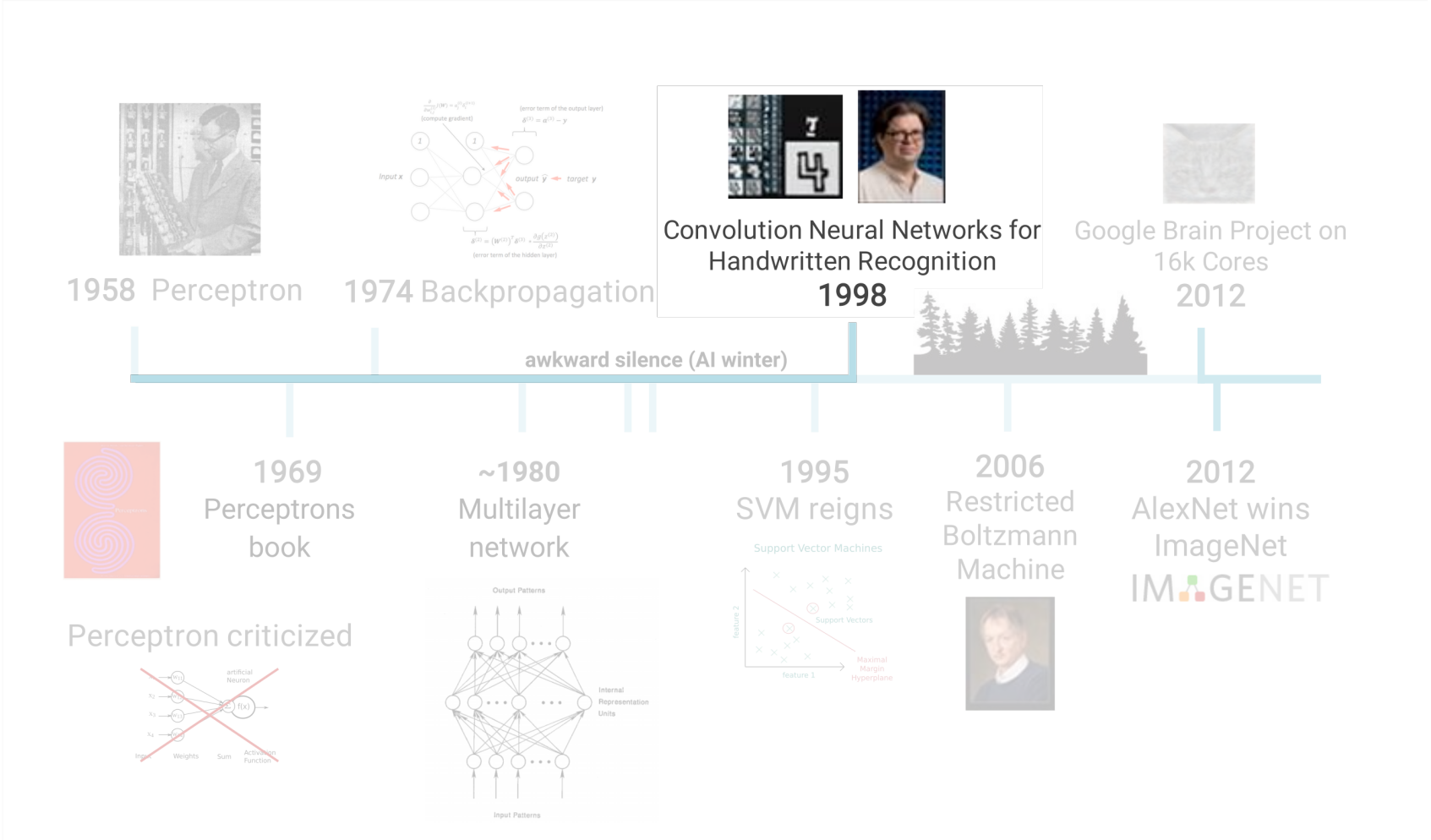
Convolutional neural networks

Timeline – Convolutional neural networks

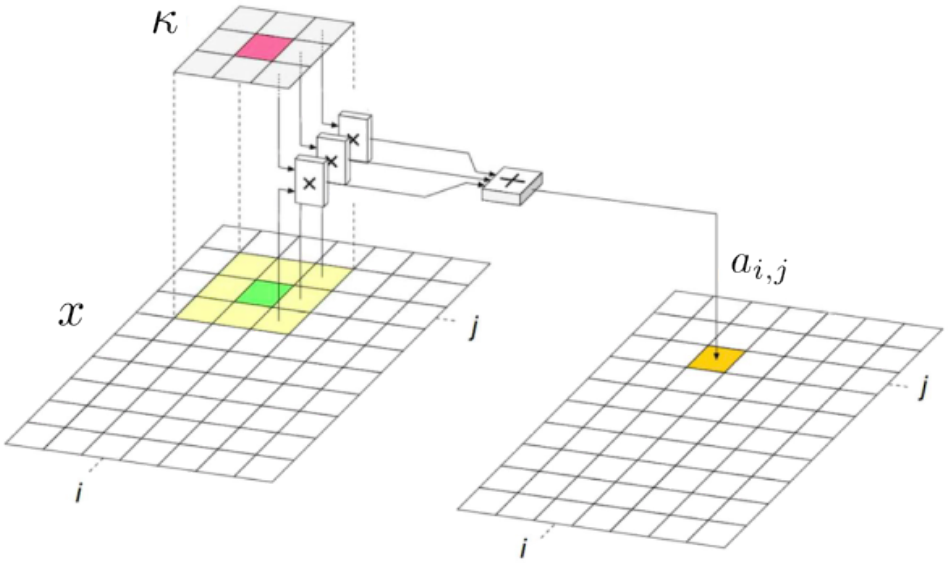
Convolutional neural networks



Linear filters: 2D convolutions

- Convolution product $I * \kappa$ of an image I by a filter κ (also called kernel):

$$(I * \kappa)(i,j) = \sum_{u=-\infty}^{+\infty} \sum_{v=-\infty}^{+\infty} I(i+u, j+v)\, \kappa(u,v)$$



Here κ is of size $S = 3 \times 3$

$$\Rightarrow \sum_{u=-\infty}^{+\infty} \sum_{v=-\infty}^{+\infty} \equiv \sum_{u=-1}^{+1} \sum_{v=-1}^{+1}$$

κ is centered on (i,j)

Example of convolution filter

Example: average filter (*box filter*)

Average of pixels in a local neighborhood, for example with $S = 3 \times 3$:

$$\kappa = \begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & \underline{\mathbf{1/9}} & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{\mathbf{1}} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Original image

Box filter 3×3

Example of convolution filter

Example: average filter (*box filter*)

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$$\kappa = \begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & \underline{\mathbf{1/9}} & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{\mathbf{1}} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Original image

Box filter 3×3

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Example: average filter (*box filter*)

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$$\kappa = \begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & \underline{\mathbf{1/9}} & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{\mathbf{1}} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Original image

Box filter 5×5

Example of convolution filter

Example: average filter (*box filter*)

Average of pixels in a local neighborhood, for example with $S = 3 \times 3$:

$$\kappa = \begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & \underline{\mathbf{1/9}} & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{\mathbf{1}} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Original image

Box filter 7×7

Convolution step-by-step

$$\text{Filter } \kappa = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{\mathbf{1}} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

	0								

Convolution step-by-step

$$\text{Filter } \kappa = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{\mathbf{1}} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

	0	10							

Convolution step-by-step

$$\text{Filter } \kappa = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{\mathbf{1}} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

	0	10	20						

Convolution step-by-step

$$\text{Filter } \kappa = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{\mathbf{1}} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

	0	10	20	30					

Convolution step-by-step

$$\text{Filter } \kappa = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{\mathbf{1}} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

	0	10	20	30	30				

Convolution step-by-step

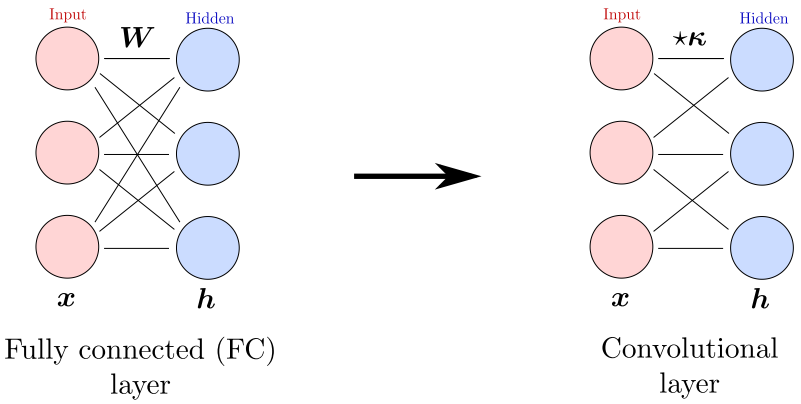
$$\text{Filter } \kappa = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{\mathbf{1}} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

	0	10	20	30	30	30	20	10	
	0	20	40	60	60	60	40	20	
	0	30	60	90	90	90	60	30	
	0	30	50	80	80	90	60	30	
	0	30	50	80	80	90	60	30	
	0	20	30	50	50	60	40	20	
	10	20	30	30	30	30	20	10	
	10	10	10	0	0	0	0	0	

What are CNNs?

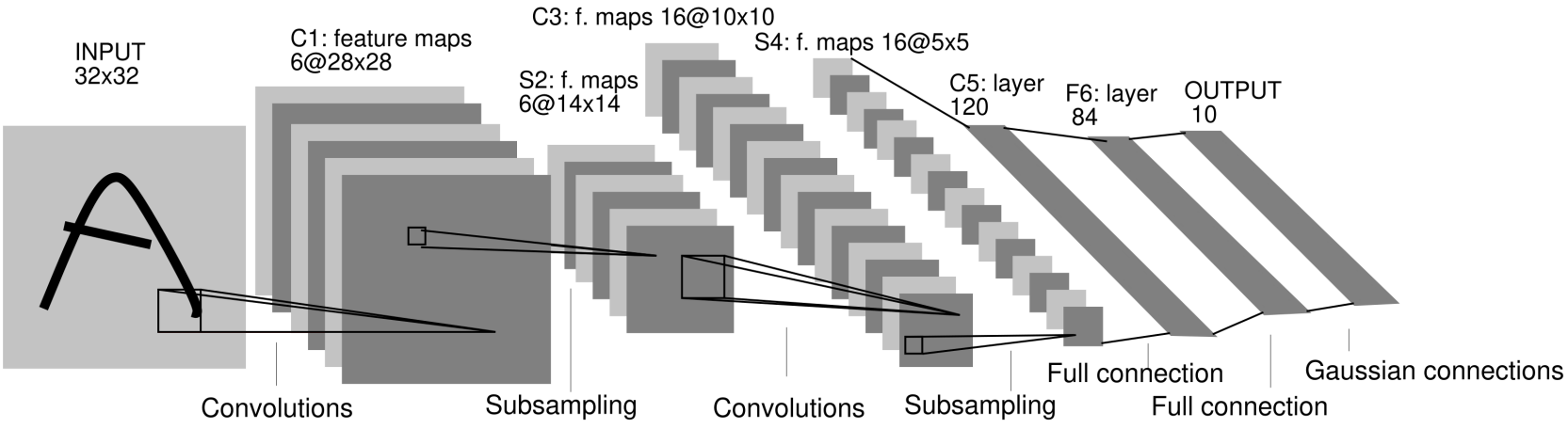
- Essentially neural networks that use **convolutions** in place of general matrix multiplications (at least for the first layers).



- CNNs are designed to process data from multidimensional arrays/tensors (*e.g.*, 2D images, 3D video or volumetric images).
- Composed of series of stages: convolutional and pooling layers.
- Do not only mimic the brain connectivity but also the visual cortex.

LeNet-5 architecture

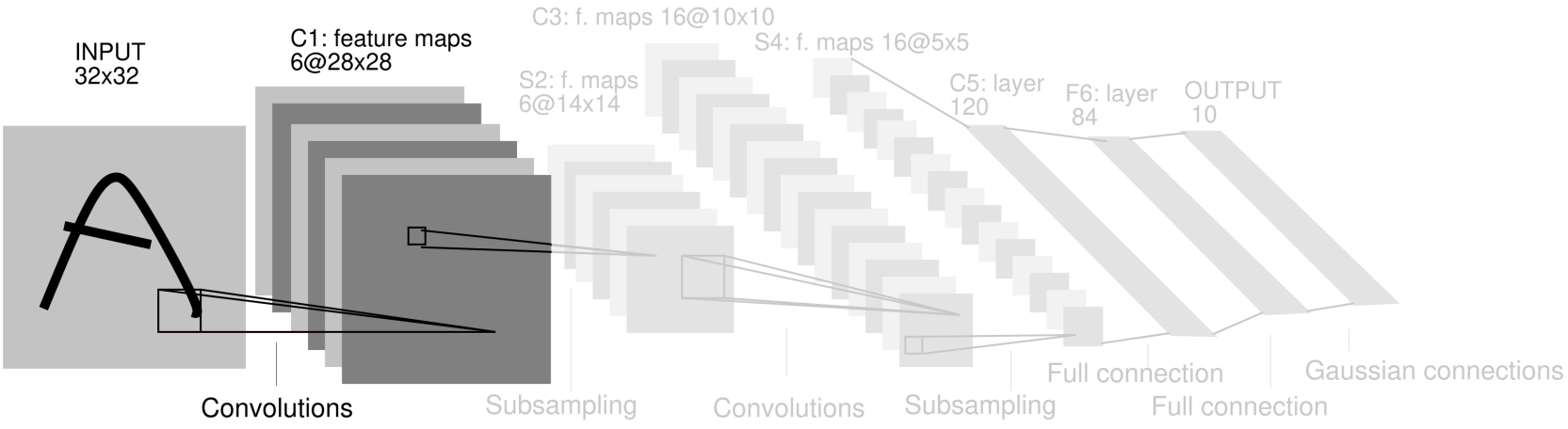
LeNet-5 (LeCun, Bottou, Bengio, Haffner, 1998)



CNN architecture for handwritten digits recognition

LeNet-5 architecture

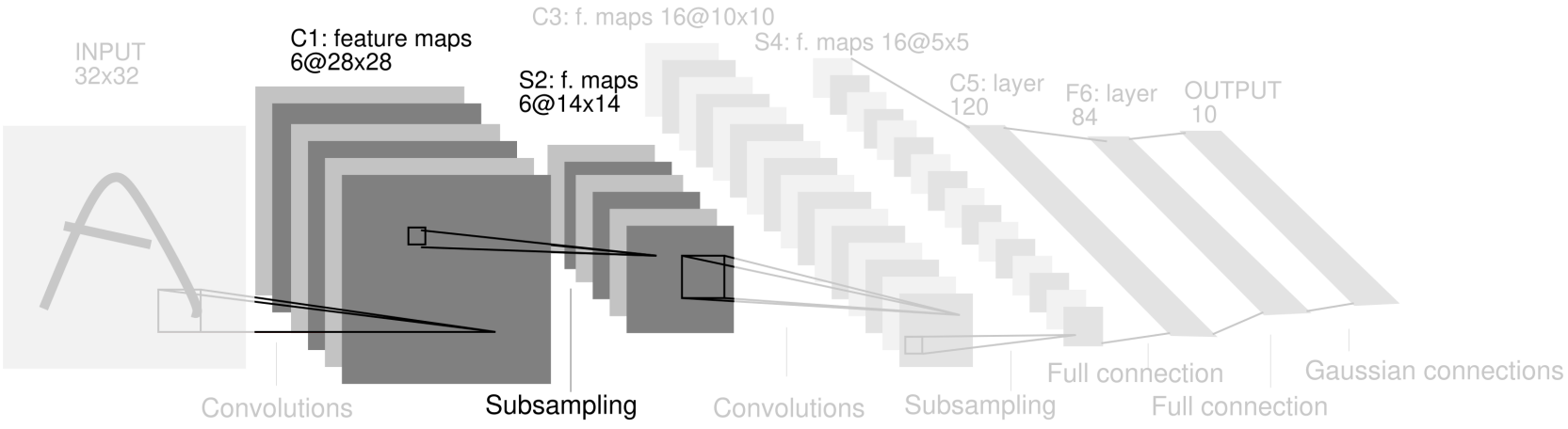
LeNet-5 (LeCun, Bottou, Bengio, Haffner, 1998)



Convolutional layer with 6 filters of size $5 \times 5\,(\times 1)$

LeNet-5 architecture

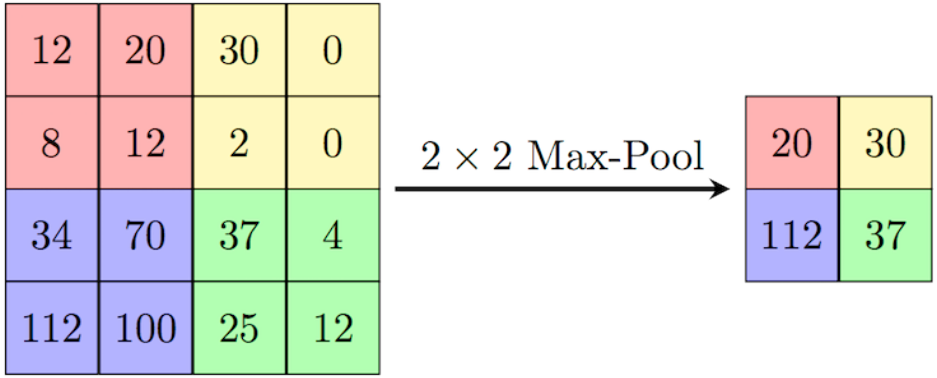
LeNet-5 (LeCun, Bottou, Bengio, Haffner, 1998)



Subsampling with 2×2 max pooling

Pooling layers

- Used after convolutional layer to aggregate information,
  - Partition the image in a grid of $z \times z$ windows (usually $z = 2$),
  - max-pooling: take the $\max$ in the window,



  - average-pooling: take the average.

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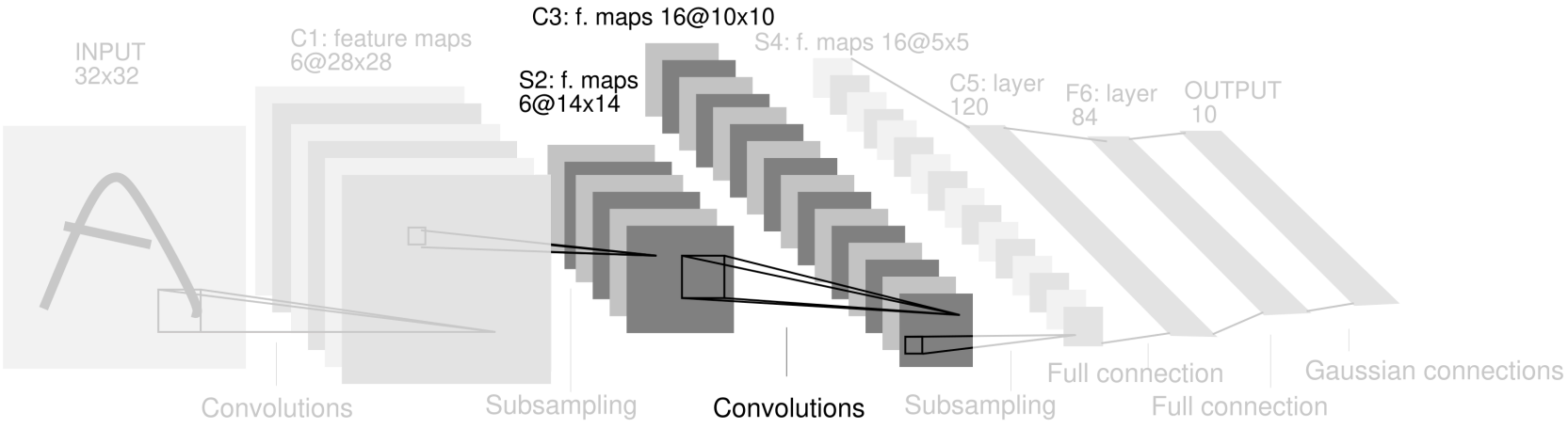
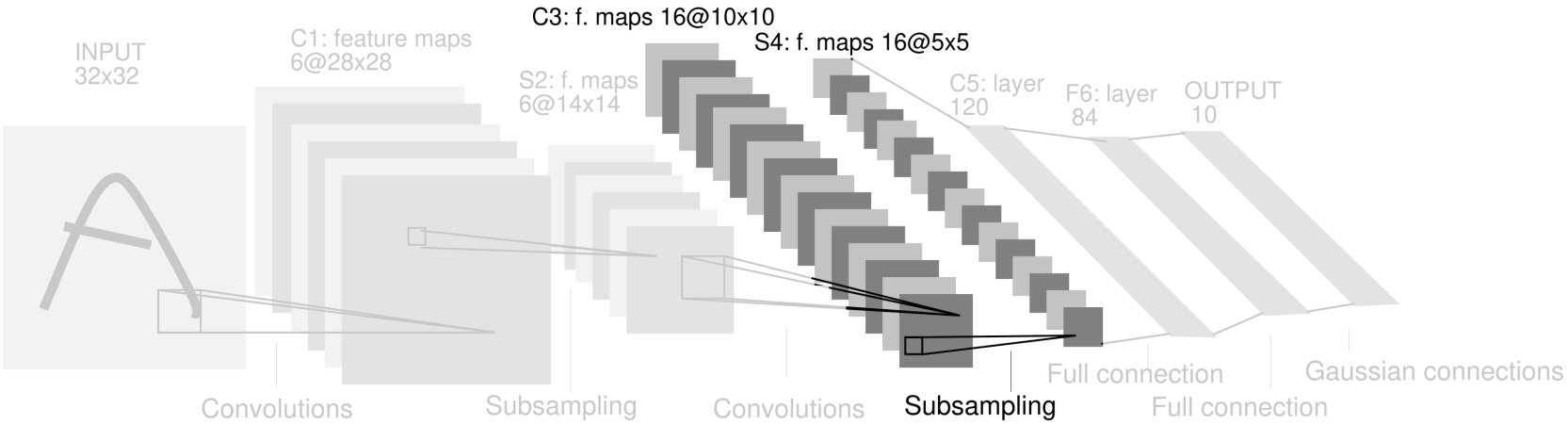


Convolutional layer with 16 filters of size $5 \times 5\,(\times 6)$

LeNet-5 architecture

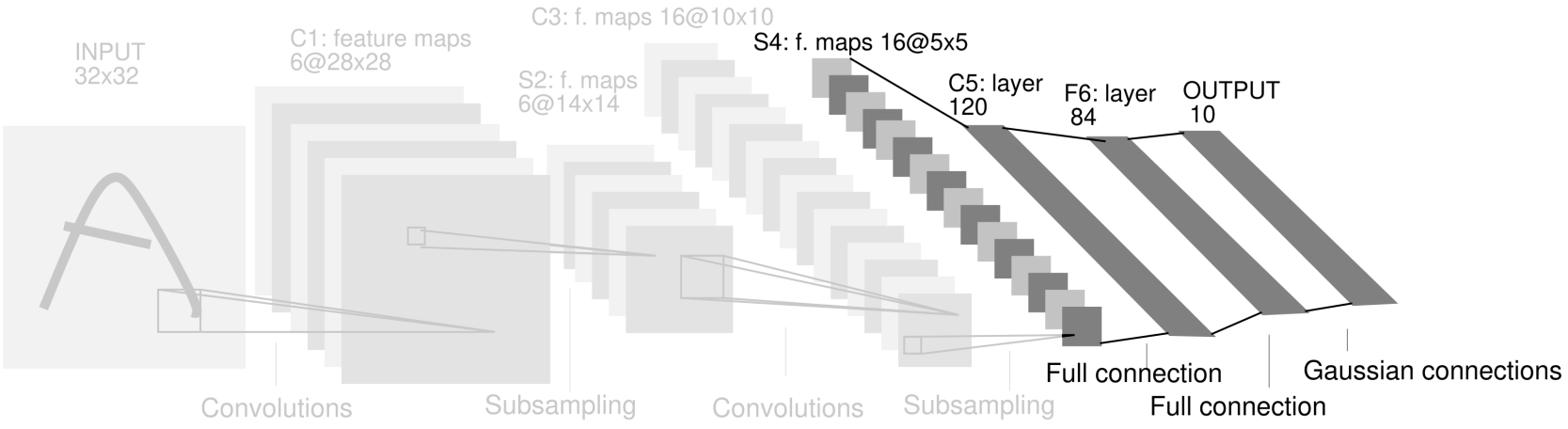
LeNet-5 (LeCun, Bottou, Bengio, Haffner, 1998)



Subsampling with 2×2 max pooling

LeNet-5 architecture

LeNet-5 (LeCun, Bottou, Bengio, Haffner, 1998)



Flattening and fully-connected layers (MLP) for classification

LeNet-5 training

LeNet-5 (LeCun, Bottou, Bengio, Haffner, 1998)

- Trained on MNIST:
  - 28×28 images
  - 60 000 for training
  - 10 000 for testing
- Used data augmentation:
  - generated random distortions
  - 600 000 training samples
- Backprop with SGD:
  - only 20 epochs
  - 60 000 parameters
  - about 3 days (at that time)

TP3 – Convolutional neural networks (CNN)